

DATABASE MANAGEMENT SYSTEM

DBMS ASSIGNMENT

COLLEGE MANAGEMENT SYSTEM

A Relational Database Design for Managing College Academic Records

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1. Introduction

A database is an organized collection of related data that can be stored, accessed, updated, and managed efficiently. In a college environment, large amounts of information are generated every day, including student details, departments, faculty records, courses, subjects, attendance, and examination marks.

The College Management System proposed in this assignment represents these real-world activities using a relational database. The design separates information into relevant tables and connects the tables using primary keys and foreign keys. This avoids unnecessary duplication and makes the data easier to maintain.

2. Name and Purpose of the Application

Application Name:

College Management System

Purpose:

The purpose of the College Management System is to store and manage core academic information about students, departments, faculty, courses, subjects, enrollment, attendance, and marks. The system provides a structured way to retrieve and update college records while maintaining relationships between related data.

3. Scope of the System

- Maintain department information.
- Store student personal and academic information.
- Maintain faculty information and department association.
- Store courses offered by each department.
- Maintain subjects and the faculty responsible for teaching them.
- Record student enrollment in courses.
- Record student attendance for individual subjects.
- Record internal and external examination marks.

4. Required Database Tables

The following eight tables are sufficient for the selected application. Each table represents a meaningful part of the college management process; no table has been added merely to increase the table count.

No.	Table	Purpose
1	Department	Stores college department information.
2	Student	Stores student personal and department information.
3	Faculty	Stores faculty information and department association.
4	Course	Stores courses offered by departments.
5	Subject	Stores subjects belonging to courses and their assigned faculty.
6	Enrollment	Records which students are enrolled in which courses.
7	Attendance	Records attendance of students for subjects.
8	Marks	Stores examination marks obtained by students in subjects.

5. Detailed Table Structures

Department Table

Column / Attribute	Data Type	Key	Description
department_id	INT	Primary Key	Unique identifier for a department
department_name	VARCHAR(100)	—	Name of the department
hod_name	VARCHAR(100)	—	Name of the Head of Department

Student Table

Column / Attribute	Data Type	Key	Description
student_id	INT	Primary Key	Unique identifier for a student
student_name	VARCHAR(100)	—	Full name of the student
email	VARCHAR(100)	—	Student email address
phone	VARCHAR(15)	—	Student contact number
date_of_birth	DATE	—	Student date of birth
department_id	INT	Foreign Key	References Department

Faculty Table

Column / Attribute	Data Type	Key	Description
faculty_id	INT	Primary Key	Unique identifier for a faculty member
faculty_name	VARCHAR(100)	—	Full name of the faculty

			member
email	VARCHAR(100)	—	Faculty email address
phone	VARCHAR(15)	—	Faculty contact number
department_id	INT	Foreign Key	References Department

Course Table

Column / Attribute	Data Type	Key	Description
course_id	INT	Primary Key	Unique identifier for a course
course_name	VARCHAR(100)	—	Name of the course/program
duration_years	INT	—	Course duration in years
department_id	INT	Foreign Key	References Department

Subject Table

Column / Attribute	Data Type	Key	Description
subject_id	INT	Primary Key	Unique identifier for a subject
subject_name	VARCHAR(100)	—	Name of the subject
credits	INT	—	Credit value of the subject
course_id	INT	Foreign Key	References Course
faculty_id	INT	Foreign Key	References Faculty

Enrollment Table

Column / Attribute	Data Type	Key	Description
enrollment_id	INT	Primary Key	Unique enrollment record
student_id	INT	Foreign Key	References Student
course_id	INT	Foreign Key	References Course
enrollment_date	DATE	—	Date of enrollment

Attendance Table

Column / Attribute	Data Type	Key	Description
attendance_id	INT	Primary Key	Unique attendance record
student_id	INT	Foreign Key	References Student
subject_id	INT	Foreign Key	References Subject
attendance_date	DATE	—	Date on which attendance is recorded
status	VARCHAR(10)	—	Present or Absent

Marks Table

Column / Attribute	Data Type	Key	Description
mark_id	INT	Primary Key	Unique marks record
student_id	INT	Foreign Key	References Student
subject_id	INT	Foreign Key	References Subject
internal_marks	INT	—	Internal assessment marks
external_marks	INT	—	External examination

			marks
total_marks	INT	—	Total marks obtained

6. Relationships Between the Tables

Relationship	Cardinality	Explanation
Department → Student	One-to-Many (1:N)	One department can have many students; each student belongs to a department.
Department → Faculty	One-to-Many (1:N)	One department can have many faculty members.
Department → Course	One-to-Many (1:N)	One department can offer multiple courses.
Course → Subject	One-to-Many (1:N)	One course can contain multiple subjects.
Faculty → Subject	One-to-Many (1:N)	One faculty member can teach multiple subjects in this simplified model.
Student → Enrollment	One-to-Many (1:N)	One student can have enrollment records.
Course → Enrollment	One-to-Many (1:N)	One course can have many student enrollment records.
Student → Attendance	One-to-Many (1:N)	One student can have many attendance records.
Subject → Attendance	One-to-Many (1:N)	One subject can have attendance records for many students.
Student → Marks	One-to-Many (1:N)	One student can have marks records for multiple subjects.
Subject → Marks	One-to-Many (1:N)	One subject can have marks records for multiple students.

7. Entity Relationship Diagram (ER Diagram)

The ER diagram below shows the main entities and the foreign-key relationships between them. PK means Primary Key and FK means Foreign Key.

	1 : N	DEPARTMENT PK department_id department_name hod_name	1 : N	
STUDENT PK student_id student_name email phone DOB FK department_id	Dept 1:N	FACULTY PK faculty_id faculty_name email phone FK department_id	Dept 1:N	COURSE PK course_id course_name duration_years FK department_id
Student 1:N	Course 1:N	Course 1:N	Faculty 1:N	Subject 1:N
ENROLLMENT PK enrollment_id FK student_id FK course_id enrollment_date	Student ↔ Course	SUBJECT PK subject_id subject_name credits FK course_id FK faculty_id	Subject 1:N	ATTENDANCE PK attendance_id FK student_id FK subject_id attendance_date status
	Student/Subject	MARKS PK mark_id FK student_id FK subject_id internal_marks	Student/Subject	

		external_marks total_marks		
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Note: The Enrollment table acts as an associative table between Student and Course, allowing the database to record which student is enrolled in which course.

8. Primary Keys and Foreign Keys

Table	Primary Key	Foreign Key(s)
Department	department_id	—
Student	student_id	department_id → Department.department_id
Faculty	faculty_id	department_id → Department.department_id
Course	course_id	department_id → Department.department_id
Subject	subject_id	course_id → Course.course_id; faculty_id → Faculty.faculty_id
Enrollment	enrollment_id	student_id → Student.student_id; course_id → Course.course_id
Attendance	attendance_id	student_id → Student.student_id; subject_id → Subject.subject_id
Marks	mark_id	student_id → Student.student_id; subject_id → Subject.subject_id

A primary key uniquely identifies each row in a table. A foreign key stores a value that refers to a primary key in another table, thereby establishing a relationship between the tables.

9. Sample Data

Department

department_id	department_name	hod_name
1	BCA – AI & ML	Dr. Kumar
2	BBA	Dr. Priya
3	MBA	Dr. Ravi

Student

student_id	student_name	email	department_id
101	Akil	akil@example.com	1
102	Rahul	rahul@example.com	1
103	Priya	priya@example.com	2

Course

course_id	course_name	duration_years	department_id
201	BCA	3	1
202	BBA	3	2
203	MBA	2	3

Subject

subject_id	subject_name	credits	course_id	faculty_id
501	Database Management System	4	201	301

502	Python Programming	4	201	302
503	Mathematics	3	201	303

10. Example Database Operations

The database can support common college operations such as finding a student's department, checking attendance, and retrieving marks.

Find all students in the BCA course

```
SELECT s.student_id, s.student_name
FROM Student s
JOIN Enrollment e ON s.student_id = e.student_id
JOIN Course c ON e.course_id = c.course_id
WHERE c.course_name = 'BCA';
```

Find marks of a student

```
SELECT s.student_name, sub.subject_name, m.total_marks
FROM Marks m
JOIN Student s ON m.student_id = s.student_id
JOIN Subject sub ON m.subject_id = sub.subject_id
WHERE s.student_id = 101;
```

Find subjects taught by a faculty member

```
SELECT f.faculty_name, sub.subject_name
FROM Faculty f
JOIN Subject sub ON f.faculty_id = sub.faculty_id
WHERE f.faculty_id = 301;
```

11. Advantages of the Proposed Database

- Reduces unnecessary duplication of information.
- Provides organized storage for academic records.
- Primary and foreign keys maintain relationships between records.
- Makes searching and retrieving information faster and easier.
- Attendance and marks can be linked directly to students and subjects.
- The structure can be expanded later without redesigning the whole system.
- Separating data into related tables supports better data consistency and maintenance.

12. Conclusion

The College Management System is a practical example of how a real-world application can be represented using a relational database. The proposed design uses eight relevant tables—Department, Student, Faculty, Course, Subject, Enrollment, Attendance, and Marks. Primary keys uniquely identify records, while foreign keys connect related information. The design provides a clear and structured way to manage essential academic data while avoiding unnecessary tables. It can also serve as a foundation for a larger college information system.